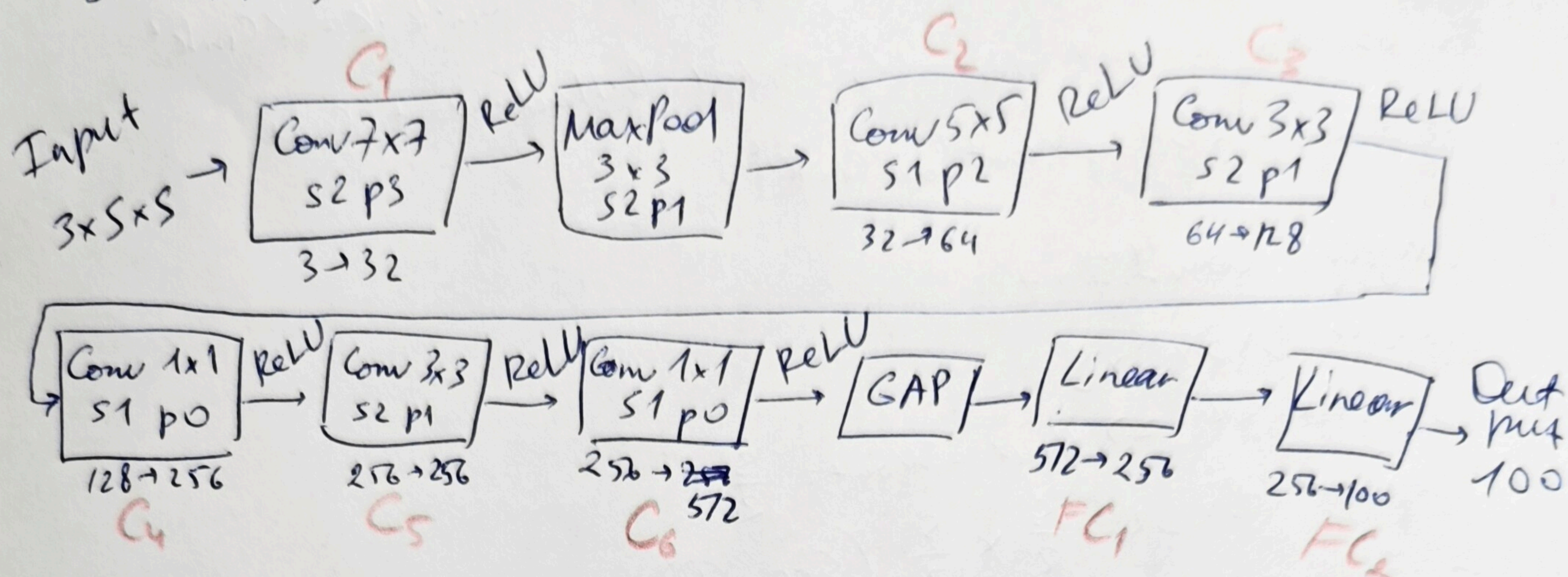


Efficient Models 4/w 1

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Tensor shapes per layer:

① $X: B \times 3 \times S \times S$

$C_1: B \times 32 \times \frac{S}{2} \times \frac{S}{2}$

$C_2: B \times 32 \times \frac{S}{4} \times \frac{S}{4}$

$C_3: B \times 128 \times \frac{S}{8} \times \frac{S}{8}$

$C_4: B \times 256 \times \frac{S}{16} \times \frac{S}{16}$

$C_5: B \times 256 \times \frac{S}{16} \times \frac{S}{16}$

$C_6: B \times 512 \times \frac{S}{16} \times \frac{S}{16}$

GAP: $B \times 512$

FC1: $B \times 256$

FC2: $B \times 100 \rightarrow \text{output}$

We assume that each weight is stored in FP32 (4 bytes per value)

Also: no Conv/Linear bias, inplace ReLU, no Batch Norm

Pooling is not taken into account in FLOPS calculations.

Parameters per layer

$W_{C1} = 3 \cdot 32 \cdot 7^2 = 4704$

$W_{C2} = 32 \cdot 64 \cdot 5^2 = 51200$

$W_{C3} = 64 \cdot 128 \cdot 3^2 = 73728$

$W_{C4} = 128 \cdot 256 \cdot 3^2 = 32768$

$W_{C5} = 256 \cdot 256 \cdot 3^2 = 589824$

$W_{C6} = 256 \cdot 512 \cdot 3^2 = 131072$

$W_{FC1} = 512 \cdot 256 = 131072$

$W_{FC2} = 256 \cdot 100 = 25600$

$\sum N_W = 4704 + 51200 + 73728 + 32768 + 589824 + 131072 + 25600 = 1039968$

$N_W = 1039968$

Each parameter is in FP32 $\Rightarrow M_W = 1039968 \cdot 4 = 4159872 \text{ bytes}$

FLOPs per 1 convolution element $2 \cdot C_{in} \cdot k \cdot k = 2C_{in}k^2$

Convolution output elements total $B \cdot H_{out} \cdot W_{out} \cdot C_{out}$

$$\Rightarrow F_{conv} = 2B H_{out} W_{out} C_{out} C_{in} k^2 \quad F_{conv} =$$

1 Layer Conv 7×7 :

input $B \times 3 \times 5 \times 5$

stride = 2 $\Rightarrow H_{out} = W_{out} = \frac{5}{2}$

Output $B \times 32 \times \frac{5}{2} \times \frac{5}{2}$ (see ①)

$$F_{C1} = \underbrace{2 \cdot 3 \cdot 7^2}_{2C_{in}k^2} \cdot B \cdot 32 \cdot \frac{5}{2} \cdot \frac{5}{2} = 294 \cdot 8BS^2 = 2352BS^2$$

$$F_{C2} = 2B \left(\frac{5}{4}\right)^2 \cdot 64 \cdot 32 \cdot 5^2 = 6400BS^2$$

$$F_{C3} = 2 \cdot 64 \cdot 32^2 \cdot B \cdot 128 \left(\frac{5}{8}\right)^2 = 2304BS^2$$

$$F_{C4} = 1024BS^2$$

$$F_{C5} = 4608BS^2$$

$$F_{C6} = 1024BS^2$$

$$\begin{aligned} \sum F_{conv} &= (2352 + 6400 + 2304 + 1024 + 4608 + 1024)BS^2 \\ &= 17712BS^2 \end{aligned}$$

FLOPs per Linear Layer:
(512 + 256)

$$F_{FC1} = 2B \cdot 512 \cdot 256 = 262144B$$

$$F_{FC2} = 2B \cdot 256 \cdot 100 = 51200B$$

$$\sum F_{linear} = 313344B$$

Total FLOPs:

$$F(S, B) = F_{conv} + F_{FC} = 17712BS^2 + 313344B$$

$$F(S, B) = B(17712S^2 + 313344)$$

Memory

ReLU is made inplace
 $\rightarrow$ no activation memory

$$M = M_w + M_A$$

$$N_A = B(3S^2 + 8S^2 + 2S^2 + 4S^2 + 2S^2 + 4S^2 + S^2 + 2S^2 + 512 + 256 + 100) =$$

$$= B(26S^2 + 868)$$

$$M_A = 4 \cdot N_A \quad (\text{FP32} = 4 \text{ bytes})$$

$$M_A = 4B(26S^2 + 868) = 104BS^2 + 3472B$$

$$M = M_w + M_A = 4159872 + 104BS^2 + 3472B \quad M(S, B)$$

Bytes moved i/o

For Conv / Linear layers

$$Q_i = N_{in} + N_w + N_{out} \quad \rightarrow \text{write}$$

$$Q_{\text{ReLU}} = N_A + N_A$$

inplace read write

$$Q_{\text{pool}} = N_{in} + N_{out}$$

read write

$$Q_{C1} = 3BS^2 + 4704 + 9BS^2 = 11BS^2 + 4704$$

$$Q_{\text{maxpool}} = 8BS^2 + 2BS^2 = 10BS^2$$

$$Q_{R1} = 2 \cdot 8BS^2 = 16BS^2$$

$$Q_{C2} = 2BS^2 + 5120 + 4BS^2 = 6BS^2 + 5120$$

$$Q_{C3} = 6BS^2 + 73728$$

$N_{in} + N_{out}$

$$Q_{R3} = 4BS^2$$

$$Q_{R2} = 8BS^2$$

$$Q_{C4} = 6BS^2 + 32768$$

$$Q_{R4} = 8BS^2$$

$$Q_{C5} = 5BS^2 + 589824$$

$$Q_{R5} = 2BS^2$$

$$Q_{C6} = 3BS^2 + 131072$$

$$Q_{R6} = 4BS^2$$

$$Q_{\text{GAP}} = 2BS^2 + 512B$$

$$Q_{\text{FC1}} = 768B + 131072$$

$$Q_{\text{FC2}} = Q_{R7} = 512B$$

$$Q_{\text{FC2}} = 356B + 25600$$

$$\sum Q_i = 1039968 + B(91S^2 + 2148) \quad \text{values in FP32}$$

N_w

$$\Rightarrow Q(S, B) = 4 \sum Q_i = 4159872 + 364BS^2 + 8592B$$

Latency

$$T(S, B, \theta) = t_0 + \max\left(\frac{F(S, B)}{P}, \frac{Q(S, B)}{BW}\right)$$

P - compute throughput

BW - memory bandwidth

t_0 - fixed overhead

Final

Fitted values after training (can be found in results/theta.json)

$$T(S, B) = 0,00040245 + \max\left(\frac{B(17712S^2 + 313344)}{3,1330 \cdot 10^{12}}, \frac{4(1039968 + B(915^2 + 2148))}{7,7074 \cdot 10^{10}}\right)$$

Energy

$$E = e_0 + e_F F_{\text{FLOP}} + e_Q Q_{\text{GB}}$$

Final formula after calibration of parameters:

$$E(S, B) = 0,01977984 + 1,242 \cdot 10^{-8} \frac{F(S, B)}{10^9} + 1,021067 \frac{Q(S, B)}{10^9} \text{ (J)} =$$

$$= 1,2 \cdot 10^{-2} + 1,242 \cdot 10^{-8} \frac{B(17712S^2 + 313344)}{10^9} + 1,021067 \frac{4(1039968 + B(915^2 + 2148))}{10^9}$$